USN [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] **[22BS2MA01]**

SECOND SEMESTER B.Tech EXAMINATION, APRIL - 2023
PROBABILITY AND VECTOR SPACES

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ii. Answer All Questions

		PART-A			05 X 02=10
1.	**a.**	Define Sample space with an example.	L-1	CO1	**02**
	b.	Briefly explain types of Random variables.	L-2	CO2	**02**
	c.	Let $V = R^3$ the vector space of all ordered triplets of real numbers, over the field of real numbers. Show that the subset $w = \{(x, 0,0)/x \in R\}$ is a subspace of R^3.	L-2	CO3	**02**
	d.	Explain linear transformation.	L-1	CO4	**02**
	e.	Find the Eigen values of the matrix $\begin{pmatrix} 1 & 6 \\ 5 & 2 \end{pmatrix}$.	L-2	CO5	**02**
		PART-B			**06 X 05=30**
2.	**a.**	When two dice are thrown simultaneously. What is the probability that the sum of the numbers in the two faces is **i.** an odd number. **ii.** an even number.	L-2	CO1	**05**
		OR			
	b.	State and prove Baye's theorem.	L-1	CO1	**05**
3.	**a.**	From a sealed box containing a dozen apples it was found that 3 apples are perished. Obtain the probability distribution of the number of perished apples when 2 apples are drawn at random. Also find mean and variance of the distribution.	L-2	CO2	**05**
		OR			
	b.	If x is an exponential variate with mean 3, then find i) $P(x > 1)$ ii) $P(x < 3)$.	L-2	CO2	**05**
4.	**a.**	Let $S = (x,\ 2x,\ -3x,\ x)$ in V_4, show that S is a subspace of V_4.	L-1	CO3	**05**
		OR			
	b.	Show that the subset $w = \{(x_1, x_2, x_3)/x_1 + x_2 + x_3 = 0\}$ of the vector space $V_3(R)$ is a subspace of $V_3(R)$.	L-1	CO3	**05**

5.	**a.**	Show that $\{v_1, v_2, v_3\}$ is an orthonormal set of R^3 where $v_1 = \begin{pmatrix} \frac{3}{\sqrt{11}} \\ \frac{1}{\sqrt{11}} \\ \frac{1}{\sqrt{11}} \end{pmatrix}$, $v_2 = \begin{pmatrix} \frac{-1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{pmatrix}$, $v_3 = \begin{pmatrix} \frac{-1}{\sqrt{66}} \\ \frac{-4}{\sqrt{66}} \\ \frac{7}{\sqrt{66}} \end{pmatrix}$.	L-3	CO4	**05**
		OR			
	b.	Let W be the three-dimensional subspace of R^3 with orthonormal basis $W = \{w_1, w_2\}$ where $w_1 = \left(\frac{2}{3}, -\frac{1}{3}, -\frac{2}{3}\right)$, $w_2 = \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)$. Using standard inner product of R^3, find the orthogonal projection of $v = (2,1,3)$ on W.	L-3	CO4	**05**
6.	**a.**	Find the Eigen values of matrix $A = \begin{bmatrix} 2 & -3 & 0 \\ 2 & -5 & 0 \\ 0 & 0 & 3 \end{bmatrix}$.	L-2	CO5	**05**
		OR			
	b.	Let $A = \begin{pmatrix} 1 & 6 \\ 5 & 2 \end{pmatrix}$, $u = \begin{pmatrix} 6 \\ -5 \end{pmatrix}$ and $v = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$. Are u and v eigen vectors of A?	L-2	CO5	**05**
7.	**a.**	Discuss python code to find $i)\ P(X = 5)$, $ii)\ P(X \geq 7)$, $iii)\ P(1 \leq x \leq 6)$ using binomial distribution where mean is 3 and variance is 9/4 respectively.	L-3	CO2	**05**
		OR			
	b.	Explain python code to verify if the two vectors $U = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$ and $V = \begin{bmatrix} 5 \\ -7 \end{bmatrix}$ for some scalar $k = -2$ satisfies conditions of linear transformation.	L-3	CO4	**05**
		PART-C	**06 X 10=60**		
8.	**a.**	Define Probability using classical, frequency and axiomatic approaches. Two aero planes bomb a target in succession. The probability of each correctly scoring a hit is 0.3 and 0.2 respectively. The second plane will bomb only if the first misses the target. Find the probability that **i.** The target is hit **ii.** Both fail to score hits.	L-3	CO1	**10**
		OR			

	b.	Define Independent event and conditional probability with suitable examples. Three major parties A, B and C are contending for power in the elections of a state and the chance of their winning the election is in the ratio 1:3:5. The parties A, B and C respectively have probabilities of banning the online lottery 2/3, 1/3 and 3/5. What is the probability that there will be a ban on the online lottery in the state? What is the probability that the ban is from the party C?	L-3	CO1	**10**
9.	**a.**	Ten coins are tossed simultaneously. Find the probability of getting **i.** At least seven heads **ii.** Exactly seven heads **iii.** At most seven heads	L-3	CO2	**10**
		OR			
	b.	The weight of ball bearings proved in a certain factory are normally distributed with mean 0.5 Newtons and standard deviation 0.002 Newtons. Find the percentage of ball bearings produced with weights. **i.** Less than 0.498 Newtons **ii.** More than 0.501 Newtons **iii.** Between 0.4995 and 0.5005 Newtons $[\emptyset(1) = 0.3413, \emptyset(0.5) = 0.1915 \; and \; \emptyset(0.25) = 0.0987]$	L-3	CO2	**10**
10.	**a.**	Let V be the set of all positive real numbers where addition is defined by $x + y = xy$ and scalar multiplication is defined by $kx = x^k$. Determine whether V is vector space or not.	L-2	CO3	**10**
		OR			
	b.	Find the fundamental subspaces for the matrix A, where $A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 6 & 4 \\ 0 & 1 & 1 \end{bmatrix}$	L-2	CO3	**10**
11.	**a.**	Discuss Dilation, Contraction, Reflection and Rotation transformation with suitable examples.	L-2	CO4	**10**
		OR			
	b.	Consider the basis $S = \{u_1, u_2, u_3\}$ for R^3 where $u_1 = (1,1,1)$, $u_2 = (-1,0,-1)$ and $u_3 = (-1,2,3)$. Use the Gram- Schmidt process to transform S to orthonormal basis for R^3.	L-3	CO4	**10**

12.	**a.**	The following data shows atmospheric pollutants y_i (relative to an EPA standard) at half an hour intervals t_i : t_i: 1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5 y_i: -0.15, 0.24, 0.68, 1.04, 1.21, 1.15, 0.86, 0.41, -0.08 Fit a quadratic polynomial for the given that.	L-3	CO5	**10**
		OR			
	b.	Find the Eigenvalue and the corresponding eigenvectors of the matrix $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$	L-3	CO5	**10**
13.	**a.**	Discuss python code to verify whether the vectors (1,1,3), (1,2 -6) and (-1,1,-5) are linearly dependent or not ?	L-3	CO4	**10**
		OR			
	b.	Find Singular value decomposition of the matrix $A = \begin{bmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{bmatrix}$.	L-2	CO5	**10**